COMP2211 Exploring AI
Fall 2025-26 Final Exam
13/12/2025
Time Limit: 180 Minutes

Name: ______________________

Stu ID: ______________________

Instructions:

1. This exam contains 24 pages (including this cover page) and 10 questions.
2. This is a closed-book, closed-notes examination.
3. Please sign the honor code statement on page 2.
4. Please write your answers only within the space in front pages. If you write your answers on the back of a page, please make explicit indications in front page to refer to your answers.
5. The last two pages are draft paper. It may be detached from the booklet if you wish. Please do not write your answers there.
6. You can use either pen or pencil. Please make sure your writings can be clearly read. If you use pencil, your appeal during paper checking might get rejected because of this.
7. All programming codes in your answers must be written in the Python version as taught in the class.
8. For programming questions, unless otherwise stated, you are **NOT** allowed to define additional classes, helper functions and use global variables, nor other library functions.

Grade Table (for instructor & TA use only)

Question	Points	Score
1	10	
2	15	
3	10	
4	10	
5	10	
6	10	
7	10	
8	11	
9	9	
10	5	
Total:	100	

The HKUST Academic Honor Code

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As members of the University community, students have the responsibility to help maintain the academic reputation of HKUST in its academic endeavors.

Sanctions will be imposed if students are found to have violated the regulations governing academic integrity and honesty.

Declaration of Academic Integrity

I confirm that I have answered the questions using only materials specifically approved for use in this examination, that all the answers are my own work, and that I have not received any assistance during the examination.

Signature: ______________________________

Question 1 [10 points]

Indicate whether the following statements are true or false by putting **T** or **F** in the given table. You get 1 point for each correct answer.

Solution:

Question	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Answer	T	F	T	T	F	F	T	F	F	F

Question 2 [15 points]

(Advanced Python Programming)

2.1 (2 points)

Solution: np.dot(image[...,:], np.array([0.299, 0.587, 0.114]))

Marking scheme:

- Fully correct answer: 2 points.
- Minor mistake: 1 point.

  E.g., using mismatched dimension: np.dot(image[...,:], np.array([[0.299], [0.587], [0.114]]))
- No further partial credits.

2.2 (2 points)

Solution:

image[:, ::-1]

image[::-1, :]

Marking scheme:

- 1 point for each TODO. 2 points in total. No further partial credits.
- No point for exchanged answer: 1) image[::-1, :], 2)image[:, ::-1].
- If you use matrix multiplication to operate on the image, please show your code correctness with Python on your computer during paper checking.

2.3 (6 points)

Solution: **Marking scheme:**

- 6 points in total.
- TODO 4: 2 points

```
# 1 point;
# 1 point if directly uses image.shape[0] and
# image.shape[1] in the following loop condition,
# but the loop boundary should be considered.
height, width = image.shape
# 1 point;
```

```
    # pad_width can be ((1, 1), (1, 1));
    # 0.5 points if any wrong argument.
    padded_image = np.pad(image, pad_width=1, mode='constant',
                          constant_values=0)
```

- TODO 5: 3 points

```
# 0.5 points for each
# 0.5 points if only call 1 flip_image for each G_x and G_y
G_x = flip_image(flip_image(G_x, horiz=True), horiz=False)
G_y = flip_image(flip_image(G_y, horiz=True), horiz=False)
# 0.5 points
# minus 0.5 points if no "height, width = image.shape"
# use wrong boundary.
for i in range(height):
    for j in range(width):
        region = padded_image[i:i+3, j:j+3]   # 0.5 points
        # 0.5 points for each
        # 0.5 points for partial correct statement
        grad_x[i, j] = np.sum(region * G_x)
        grad_y[i, j] = np.sum(region * G_y)
```

- TODO 6: 1 point

```
edge_image = np.sqrt(grad_x**2 + grad_y**2)   # 1 point
```

- No further partial credits.

2.4 (5 points)

Solution: **Marking scheme:**

- 5 points in total. TODO 7:

```
# 0.5 points
# no point for random or constant initial T_old
T_old = np.mean(image)
while True:
    R_1 = image[image < T_old]    # 0.5 points
    R_2 = image[image >= T_old]   # 0.5 points
    u_1 = np.mean(R_1)    # 0.5 points
    u_2 = np.mean(R_2)    # 0.5 points
    T = 0.5 * (u_1+u_2)   # 0.5 points
    if T == T_old:
        break     # 0.5 points
```

```
            T_old = T      # 0.5 points
        thresh_image = (image >= T).astype(int)    # 0.5 points
        thresh_image = thresh_image * 255    # 0.5 points
```

- No further partial credits.
- Minus 0.5 points for C++ syntax, like using '||'.
- Minus 0.5 if not showing correct Python program syntax, like using do {...} while (cond)

Question 3 **[10 points]**
(Naïve Bayes Classifier)

3.1 (2 points)

Solution: No. Word "make" is irrelevant to either Class 0 or Class 1. If computing with Naïve Bayes Classifier without α-Laplace smoothing, both Class probabilities are 0, which are not comparable.

Marking scheme:

- 1 point for answering "No".
- 1 point for correct reason.
- No further partial credits.

3.2 (1 point)

Solution: (2), (1).
Marking scheme:
0.5 points each. 1 point in total. No further partial credits.

3.3 (a) (3 points)

Solution:
$P(w_1|Class0) = \frac{12+1}{73+10} = 0.16$
$P(w_3|Class0) = \frac{0+1}{73+10} = 0.01$
$P(w_6|Class0) = \frac{2+1}{73+10} = 0.04$

$P(w_1|Class1) = \frac{3+1}{90+10} = 0.04$
$P(w_3|Class1) = \frac{18+1}{90+10} = 0.19$
$P(w_6|Class1) = \frac{14+1}{90+10} = 0.15$

Marking scheme:

- 0.5 points for each likelihood. 3 points in total.
- If the numerator and denominator are correct, but the final answer is wrong/missing, also full mark.
- No further partial credits.

(b) (2 points)

Solution:
$P(Class\ 0|d) \propto P(Class\ 0) \cdot \prod_{k=1}^{|V|} P(w_k|Class\ 0)^{x_k} = 0.4 \cdot (0.16 \cdot 0.01 \cdot 0.04^2) \approx 1.02 \cdot 10^{-6}$
$P(Class\ 1|d) \propto P(Class\ 1) \cdot \prod_{k=1}^{|V|} P(w_k|Class\ 1)^{x_k} = 0.6 \cdot (0.04 \cdot 0.19 \cdot 0.15^2) \approx 1.03 \cdot 10^{-4}$
The final prediction is Class 1.
Marking scheme:

- 0.5 points for each correct posterior probability (numerators are enough). 1 point in total. If the multipliers are correct, but the final answer is wrong/missing, also full mark.
- The wrong use of Class Priors will get 0.25 deduction.
- 1 point for the answer "Class 1".
- No further partial credits.

(c) (1 point)

Solution: It is better to perform all computations by summing logs of probabilities rather than multiplying probabilities. Because multiplying many small probabilities can produce very tiny numbers that underflow to zero in floating-point arithmetic. Using logarithms converts products into sums of logs. This keeps numbers in a numerically stable range and avoids underflow.
Marking scheme:

- 0.5 points for the implementation method.
- 0.5 points for the reason.
- No further partial credits.

(d) (1 point)

Solution: It may decrease the accuracy. w_9 is strongly associated with Class 1, it is a discriminative feature that helps identify Class 1 emails. Removing an informative feature removes useful evidence, which can degrade classification performance, especially for documents where that feature is present.
Marking scheme:

- 0.5 points for the effect.
- 0.5 points for the reason. If only answer with less feature for training will only get 0.25 points.

- No further partial credits.

Question 4 [10 points]
(K-Nearest Neighbors)

4.1 (3 points)

Solution:

- $d(T1, P1) = |1200 - 1300| + |10 - 1| = \mathbf{109}$ *(Closest)*
- $d(T1, P2) = |1200 - 5000| + |10 - 8| = 3802$
- $d(T1, P3) = |1200 - 500| + |10 - 5| = 705$
- $d(T1, P4) = |1200 - 1000| + |10 - 10| = 200$
- $d(T1, P5) = |1200 - 4000| + |10 - 3| = 2807$

The **Loyalty Points** feature dominated the distance calculation.
Marking scheme:

- 0.5 points for each distance. 2.5 points in total.
- 0.5 points for answering "Loyalty Points".
- No further partial credits.

4.2 (1 point)

Solution: The nearest neighbor is **P1** (Distance = 109).
Prediction: **Stay** (Class of P1).
Marking scheme:

- 0.5 points for "P1" as the nearest neighbor.
- 0.5 points for "Stay" as the prediction.
- No further partial credits.

4.3 (1.5 points)

Solution: If $K = 5$, the model votes using the entire dataset (3 Leave vs. 2 Stay). It will always predict **Leave** (the majority class) regardless of input features. This is an example of **Underfitting**, because the model will perform poorly on both the training data and new data.
Marking scheme:

- 0.5 points for the answer "Leave".
- 0.5 points for the answer "Underfitting".
- 0.5 points for correct reason.
- No further partial credits.

4.4 (a) (2 points)

Solution:

	Predicted Leave	**Predicted Stay**
Actual Leave	TP = **4**	FN = **2**
Actual Stay	FP = **1**	TN = **3**

Marking scheme:
0.5 points for each entry. 2 points in total. No further partial credits.

(b) (1 point)

Solution:

$$\text{Precision} = \frac{4}{4+1} = \frac{4}{5} = \mathbf{0.8} \quad (80\%)$$

$$\text{Recall} = \frac{4}{4+2} = \frac{4}{6} \approx \mathbf{0.67} \quad (66.7\%)$$

Marking scheme:

- 0.5 points for correct precision.
- 0.5 points for correct recall.
- No further partial credits.

(c) (1.5 points)

Solution: **Recall** is more important. The problem states we must identify *every* customer who might leave (minimizing False Negatives), even at the cost of annoying some loyal customers (tolerating False Positives).
Marking scheme:

- 0.5 points for the answer "Recall".
- 1 point for fully correct reason. Deduct 0.5 points if minor mistake.
- No further partial credits.

Question 5 [10 points]
(K-Means Clustering)

5.1 (4 points)

Solution: After one iteration:

$$C_1 = \{x_1, x_2\} = \{1, 2\}, \qquad C_2 = \{x_3, x_4\} = \{5, 6\}.$$

$$\mu_1^{(1)} = 1.5, \qquad \mu_2^{(1)} = 5.5.$$

SSE $= (1-1.5)^2 + (2-1.5)^2 + (5-5.5)^2 + (6-5.5)^2 = 0.25 + 0.25 + 0.25 + 0.25 = 1.$

Marking scheme:

- 0.5 points for each x_i cluster assignment. 2 points in total.
- 0.5 points for each updated centroid. 1 point in total.
- 1 point for SSE.
- No further partial credits.

5.2 (2 points)

Solution: K-Means Clustering is sensitive to the initial centroids because:

- The algorithm only guarantees convergence to a *local* minimum of the objective function (sum of squared distances to centroids).
- Different initial centroids can lead to very different local minima, resulting in different final clusters and potentially poor solutions.

Marking scheme:

- 1 point for stating "K-Means Clustering only guarantees convergence to a *local* minimum".
- 1 point for stating "Different initial centroids can lead to very different local minima".
- Other reasonable answers, if not answering on 'local minima", will get 1 point.
- No further partial credits.

5.3 (2 points)

Solution: Two methods (only one method is enough):

(a) Run the K-Means algorithm with different centroids seeds for multiple times. The select the best model with best clustering quality.

(b) Select initial cluster centroids using sampling based on an empirical probability distribution of the points' contribution to the overall inertia.

Markings scheme:

- 1 point for proposing the method.
- 1 point for explaining the method.
- No further partial credits.

5.4 (2 points)

Solution: Quality 1: Isolation; or maximizing inter-cluster distance.
Quality 2: Compactness; or minimizing intra-cluster distance.
Marking scheme:
1 point for each quality. 2 points in total. No further partial credits.

Question 6 [10 points]
(Multilayer Perceptron)

6.1 (a) (0.5 points)

Solution: Sigmoid
Marking scheme:
0.5 points for correct answer. No partial credits.

(b) (1 point)

Solution: Training data: (720, 4), Training label: (720,) or (720, 1). Alternatively, (800, 4) / (800,) or (800, 1), if you consider yourself also inside the dataset.
Marking scheme:

- 0.5 points for training data shape.
- 0.5 points for training label shape.
- No further partial credits.

(c) (1 point)

Solution: $(4+1) \times 16 \ + \ (16+1) \times 8 \ + \ (8+1) \times 1 = 225$
Marking scheme:
1 point for correct answer. Deduct 0.5 points if minor mistake (e.g., missing biases, or correct steps but wrong answer).

6.2 (a) (1 point)

Solution: Due to random weight initialization and small datasets, the first few training batches contribute to high training loss. In contrast, the model evaluates the validation dataset after completing all training batches for that epoch, which can result in a better validation loss during the first epoch.
Marking scheme:
1 point for correct answer. Deduct 0.5 points if minor mistake/insufficient explanation.

(b) (2 points)

Solution: Overfitting. Possible methods (only two are enough):

- Collect more data

- Dropout or other regularization techniques
- Reduce model size
- Other reasonable methods to prevent overfitting

Marking scheme:

- 1 point for the cause "overfitting".
- 0.5 points for each correct method. 1 point in total.
  - Similar methods contribute 0.5 points in total.
  - "regularization" is too general. 0 points without explanation.
  - Only grade the first two if several are listed.
- No further partial credits.

(c) (1 point)

Solution: Data/class imbalance issue. Usually, the dataset has significantly more normal players than griefers. Possible solutions are oversampling the minority class or undersampling the majority class.
Marking scheme:

- 0.5 points for the cause.
- 0.5 points for the solution.
- "Small dataset" without mentioning the imbalance only got half credits.
- No further partial credits.

6.3 (a) (1 point)

Solution: The **dying ReLU problem**. Once a neuron's input is negative, it stops contributing to the learning process, and its weights are not updated during **backpropagation** because the **gradient** flowing back through the neuron becomes zero.
Marking scheme:
1 point for fully correct explanation. Deduct 0.5 points if minor mistake/insufficient explanation (e.g., not mentioning any bolded keywords).

(b) (1 point)

Solution: The suggested activation function (Leaky ReLU) allows a small portion of the input to pass through. This means that even when a neuron's input is negative, it has a gradient.
Marking scheme:
1 point for fully correct explanation. Deduct 0.5 points if minor mistake/insufficient explanation.

6.4 (1.5 points)

Solution:

- Change the output layer size to 4.
- Change the activation function to softmax.
- Change the loss function to categorical/multi-class cross-entropy loss.

Marking scheme:
0.5 points for each change. 1.5 points in total. No further partial credits.

Question 7 [10 points]
(Digital Image Processing)

7.1 (4 points)

Solution:

Operation	**Type** (Point or Local)	**Linearity** (Yes or No)
Grayscale thresholding	Point	No
Brightness shift with clipping	Point	No
Averaging filter	Local	Yes
Median filter	Local	No

Marking scheme:

0.5 points for each entry. 4 points in total. No further partial credits.

7.2 (a) (1 point)

Solution: With K_1

$$\text{sum of patch} = 10 + 10 + 10 + 10 + 50 + 50 + 10 + 50 + 50 = 250,$$

so

$$I'_1(1,1) = \frac{1}{9} \times 250 = \frac{250}{9} \approx 28 \text{ (or 27)}.$$

Marking scheme:
1 point for correct answer. No partial credits.

(b) (1 point)

Solution: With K_2,

$$\begin{aligned} I'_2(1,1) &= 5 \cdot 50 - 1 \cdot 10 - 1 \cdot 50 - 1 \cdot 10 - 1 \cdot 50 \\ &= 250 - 10 - 50 - 10 - 50 = 130. \end{aligned}$$

Marking scheme:
1 point for correct answer. No partial credits.

(c) (1.5 point)

Solution: K_1 reduces the contrast between the bright center and darker surroundings ($50 \to$ about 28), so it *blurs* the image. K_2 increases the center value ($50 \to 130$) while subtracting neighbors, enhancing edges, so it *sharpens* the image.
Marking scheme:

- 0.5 points for answering K_1 blurs.
- 0.5 points for answering K_2 sharpens.
- 0.5 points for explaining why K_1 blurs and K_2 sharpens.
- No further partial credits.

7.3 (1.5 points)

Solution: The histogram has one group of darker intensities around 50–100 (counts 40 and 60) and a brighter group around 200–250 (counts 35 and 55). Gray level 150 has a low count (10), forming a valley. Among the candidates, $T = 130$ lies closest to this valley between the modes, so $T = \boxed{130}$ is the most reasonable choice.

Marking scheme:

- 0.5 points for answering 130.
- 1 point for fully correct explanation. Deduct 0.5 points if minor mistake.

7.4 (1 point)

Solution: To improve contrast of an image.

Marking scheme:

1 point for correct answer. No partial credits.

Question 8 [11 points]
(Convolutional Neural Network)

8.1 (a) (4 points)

Solution:

Layer	**Output Shape**	**# Weights**	**# Biases**
INPUT	$(3, 10, 64, 64)$	0	0
CONV3D-1	$(16, 10, 64, 64)$	$16 \times 1 \times 5 \times 5 \times 3$	16
CONV3D-2	$(32, 10, 64, 64)$	$32 \times 3 \times 3 \times 3 \times 16$	32
POOL	$(32, 5, 32, 32)$	0	0
FLATTEN	163840	0	0
FC	10	163840×10	10

Marking scheme: 0.4 points for each entry. 4 points in total. No further partial credits.

(b) (3 points)

Solution: Step 1: FLOPs for Each Filter/Kernel
For each filter, the 3D convolution computes an output value for each position in the output volume. To compute one output value, the kernel (size $3 \times 3 \times 3$) needs to perform a dot product operation between the filter weights and the corresponding input values (i.e., the kernel is applied to a local region of the input).
Number of multiplications: For each output element, the kernel performs $3 \times 3 \times 3 = 27$ multiplications (one for each element in the kernel).
Number of additions: Each multiplication needs to be added to the accumulated sum, so there are (27 - 1 = 26) additions for each output element.
Thus, for each output element, the total number of operations (multiplications + additions) is:

$$\begin{aligned}\text{Operations per output element} &= 27 \text{ (multiplications)} + 26 \text{ (additions)} + 1 \text{ (bias)} \\ &= 54\end{aligned}$$

Step 2: Numbers of Output Elements
Now, for each of the 32 filters, we need to compute the FLOPs for all the output elements. The output size is $32 \times 10 \times 64 \times 64$, so the total number of output elements is:

$$\text{Total output elements} = 32 \times 10 \times 64 \times 64 = 1310720$$

Step 3: Total FLOPs for the Convolutional Layer Each output element requires 53 operations, so the total number of FLOPs for one filter is:

$$\text{FLOPs per filter} = 54 \times 1310720 = 70778880$$

Thus, the total number of FLOPs is: [$\boxed{70778880}$]

Marking scheme:

- 3 points for the correct answer.
- If the final answer is correct, no need to have calculation steps.
- If the final answer is wrong, 1 point for each correct step.
- No further partial credits.

8.2 (2.5 points)

Solution: TODO 1: k_h + (k_h - 1) * (d - 1)

TODO 2: k_w + (k_w - 1) * (d - 1)

TODO 3: np.pad(img, ((pad_top, pad_btm), (pad_left, pad_right)), mode='constant', constant_values=0)

TODO 4: img_pad[r: r + eff_k_h: d, c: c + eff_k_w: d]

TODO 5: np.sum(roi * kernel)

Marking scheme:

0.5 points for each TODO task. 2.5 points in total. No further partial credits.

8.3 (1.5 points)

Solution: TODO 1: np.random.rand(N, 1, 1, C) < keep_prob

TODO 2: X * mask

TODO 3: X / keep_prob

The code below are complete:

Marking scheme:

0.5 points for each TODO task. 1.5 points in total. No further partial credits.

Question 9 [9 points]
(Minimax and Alpha-beta Pruning)

9.1 (3.5 points)

Solution:

A	**B**	**C**	**D**	**E**	**F**	**G**
5	4	5	4	5	6	5

Marking scheme:
0.5 points for each entry. 3.5 points in total. No further partial credits.

9.2 (2.5 points)

Solution: A → C → G → P
A → C → H
The path A → C → H is preferable because it results in a faster victory (shorter path).
Marking scheme:

- 0.5 points for each path. 1 point in total.
- deduct 0.5 points for each wrongly added path. 0 points in minimum.
- 0.5 points for correctly stating which one is preferable.
- 1 point for a fully correct explanation.
- No further partial credits.

9.3 (2 points)

Solution: Pruned nodes version:
Left-to-right pruning: L
Right-to-left pruning: O, M, D, (I, J)
Pruned edges version:
Left-to-right pruning: E → L
Right-to-left pruning: G → O, F → M, B → D
Marking scheme:

- 0.5 points for each correct pruned node. 2 points in total.
- Deduct 0.5 points for each missing correct node or extra wrong node.

- Each node should be in the clear and correct order, e.g., L in the left-to-right order; otherwise, no count (e.g., if you write L in the right-to-left order or not indicating the order, no point for this node).
- For right-to-left pruning, if already having D in answer, no need to write I and J. If having I and/or J but missing D, deduct 0.2/0.4 points for this missing node.
- Minimum possible points: 0.

9.4 (1 point)

Solution: Node B's value should be in the range [4, 9), i.e., less than 9 and greater than or equal to 4.

Marking scheme:

0.5 points for each correct boundary condition. 1 point in total. No further partial credits.

Question 10 [5 points]
(Ethics of Artificial Intelligence)

10.1 (1 point)

Solution: Arrest data perpetuates existing social biases. For example, the data may focus on “street crimes” carried out in low-income communities of color, while neglecting other illegal activities that are carried out in more affluent and white contexts.

Marking scheme:

- 0.5 points for “bias” as the problem.
- 0.5 points for reasonable and clear explanation.
- No further partial credits.

10.2 (3 points)

Solution: Arrest data in history records primarily reflect law enforcement activity and the deeper historical disparities in how the police and court officials treat different groups and pursue various types of crime. Such data perpetuates existing social biases.

After COMPAS was trained on this type of historical data, the model would more easily predict the marginalized group as risky recidivist, who then have a high chance of pretrial arrest.

Then it generated more biased training data for COMPAS, which further reinforces the bias and creates a negative feedback loop of increasing the existing marginalization.

Marking scheme:

- 1 point for analyzing “such data perpetuates existing social biases”.
- 1 point for analyzing “the model would more easily predict the marginalized group as risky recidivist if trained on the biased data”.
- 1 point for analyzing “the model generated more biased training data”.
- No further partial credits.

10.3 (1 point)

Solution: No. Even if protected features are not used by model, there may be other proxy features that are correlated with protected features.

Marking scheme:

- 0.5 points for the answer “No”.
- 0.5 points for correct reason.
- No further partial credits.